

Implicit Camera Calibration Using MultiLayer Perceptron Type Neural Network

Dong-Min Woo and Dong-Chul Park
Image Processing Lab
Department of Information Engineering
Myongji University, Gyeonggi-do, Korea 449-728
dmwoo@mju.ac.kr

Abstract

This paper suggest a new camera calibration approach based on the neural network model. The proposed approach is shown to be very accurate because the neural network model implicitly contains all the physical parameters, some of which are very difficult to be estimated in the conventional explicit calibration methods. As the first step of this approach, this paper presents the camera calibration process which enables the coordinate transformation between 2D image points and points of a certain space in 3D real world. However, this approach is currently extended to be a general 3D camera calibration method in terms of 2 plane method. Experimental comparison of our method with well-known Tsai's 2 stage method is made to verify the accuracy of the proposed method.

1. Introduction

Camera calibration is a process to obtain the correlation between camera image coordinate and 3D real world coordinate. Most calibration methods are based on the camera model which consists of physical parameters of the camera including position, orientation, focal length, and optical center [1, 2, 6]. However, physical parameters of a camera are not necessarily available in some stereo vision cases. In this case, we have to use an implicit calibration method. In particular, when the lens distortion is excessive and the image center is assumed to be the center of the frame grabber, it may be difficult to align both the CCD cells and lens in a perfectly parallel position. Some intermediate parameters should be calibrated by estimating image coordinates from known world coordinates. Martins first proposed the two-plane method [3]. Martins' two-plane method considers lens distortions. However, in general, calibrated parameters are not globally valid in the whole image plane. The more recent work [4] can be used for both 3D reconstruction and the computation of image coordinates. However,

lens distortion has not been considered in the implicit camera calibration approach.

In this paper, a new camera calibration approach based on an artificial neural network(ANN) model is proposed. ANNs have been shown to have the ability to model an unspecified nonlinear relationship between input patterns and output patterns. This nonlinear mapping ability can be utilized to address some physical parameters in implicit camera calibration that cannot be readily estimated by the existing calibration methods. The ANN-based camera calibration approach does not estimate camera physical parameters. However, this is not an issue when the objective of the camera calibration process is to obtain the correlation between the camera image coordinates and the 3D real world coordinates. The implicit camera calibration approach, which can calibrate a camera without explicitly computing its physical parameters, can be used for both the 3D measurement and the generation of image coordinates.

2. Explicit Camera Calibration

Tsai's two stage method (TSM)[6], which is considered one of the most powerful explicit methods based on pinhole camera model, is chosen as a reference state of art method for the purpose of performance comparison. The TSM first obtains the transformation parameters with the assumption that there exists no distortion in the camera. The TSM then refines the transformation parameters with the distortion of the camera by using a nonlinear search. That is, first, the camera model is assumed to be ideal for the camera calibration by neglecting the lens distortion.

Fig. 1 shows the camera model used in TSM. A point P is an object of the real world coordinate(X_w, Y_w, Z_w) and (x, y, z) is a 3D camera coordinate. The center of the camera coordinate is the optical center O and (X, Y) is the image coordinate with the center of O_i . The distance between O and O_i is f , the focal length of the camera. (X_u, Y_u) is the corresponding point with the assumption of no lens distortion. (X_u, Y_u) is then translated to (X_f, Y_f), which is a point in

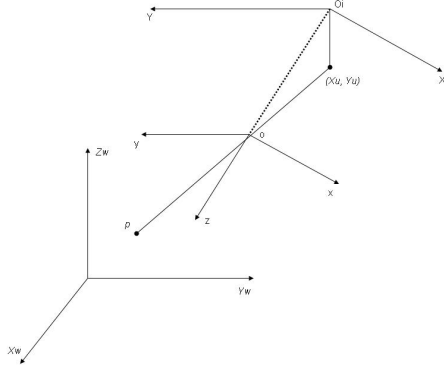


Figure 1. Pinhole camera model.

computer image coordinate on the image buffer and is expressed in pixel numbers. The basic geometry of the camera model can be written as the transformation of the two coordinates with the following displacement and orientation:

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ r_7 & r_8 & r_9 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} \quad (1)$$

with

$$\begin{aligned} r_1 &= \cos \psi \cos \theta \\ r_2 &= \sin \psi \cos \theta \\ r_3 &= -\sin \theta \\ r_4 &= -\sin \psi \cos \theta + \cos \psi \sin \theta \cos \phi \\ r_5 &= \cos \psi \cos \theta + \sin \psi \sin \theta \sin \phi \\ r_6 &= \cos \theta \sin \phi \\ r_7 &= \sin \psi \sin \phi + \cos \psi \sin \theta \cos \phi \\ r_8 &= -\cos \psi \sin \phi + \sin \psi \sin \theta \cos \phi \\ r_9 &= \cos \theta \cos \phi \end{aligned}$$

where θ , ϕ and ψ represent yaw, pitch and roll, respectively.

As can be seen from the above equations, there are six extrinsic parameters: θ , ϕ , and ψ for rotation, and three components for the translation vector T . The problem of camera calibration is to find the six parameters θ , ϕ , ψ , T_x , T_y , and T_z by using the number of points measured in the (X_w, Y_w, Z_w) coordinate.

In the second stage of the TSM, a distortion parameter is considered. The relations between the computer image coordinate with distortion and the real world coordinate can be derived as follows:

$$\begin{aligned} S_x(X_f - C_x)(1 + G(X_d^2 + Y_d^2)) \\ = f \left(\frac{r_1 x_w + r_2 y_w + r_3 z_w + T_x}{r_7 x_w + r_8 y_w + r_9 z_w + T_x} \right) \end{aligned} \quad (2)$$

$$\begin{aligned} S_x(X_f - C_x)(1 + G(X_d^2 + Y_d^2)) \\ = f \left(\frac{r_4 x_w + r_5 y_w + r_6 z_w + T_x}{r_7 x_w + r_8 y_w + r_9 z_w + T_x} \right) \end{aligned} \quad (3)$$

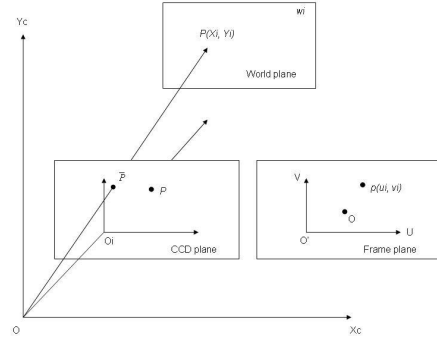


Figure 2. The imaging geometry.

where (X_f, Y_f) is the image coordinate of the frame grabber, (C_x, C_y) is the image center, S_x and S_y are components of the translating scale of the x-axis and y-axis when the A/D transform is performed, (X_d, Y_d) is a distorted coordinate by lens distortion, and G is the distortion parameter. TSM obtained the solution by using a gradient-based nonlinear search method. In an explicit calibration, the calibration is performed with extrinsic parameters. However, the distortion parameters cannot include all the parameters involved in the distortion of the image. Even with the assumption of perfect inclusion of distortion parameters, there still remains room for errors in finding the right solution for such parameters.

3. Implicit Camera Calibration

3.1. Camera Calibration Process

Suppose that there is a calibration plane and the center of the calibration plane is defined as O . In the calibration plane, we have N points. A point, $P: (X_i, Y_i) \in w_i$, $i = 1, 2, \dots, N$, in the world plane is ideally projected to $\bar{p} : (\bar{x}_i, \bar{y}_i)$ in the camera CCD plane. However, because of the distance of the camera lens, the point of the world plane is projected to a distorted point, $p: (x_i, y_i)$. This point is observed through the frame buffer coordinate $p(u_i, v_i)$ in pixels, as shown in Fig. 2.

For a back-projection problem, a transformation from the image coordinates in the frame buffer to the world coordinates in the calibration plane is required. For this purpose, an ANN is adopted in the proposed ANN-based calibration approach, where the input and the output of the ANN are the image coordinates and the world coordinates, respectively. After proper training of the ANN with training points, the ANN can map the relation of two planes. Owing to the nonlinear system modeling capability of the ANN, it is not necessary to utilize all the physical parameters involved with

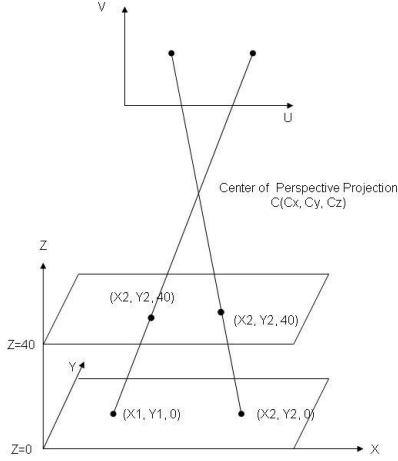


Figure 3. The center of a perspective projection.

the camera calibration, including the lens distortion and the focal length of the camera.

With the coordinate system shown in Fig. 3, $(x_1, y_1, 0)$ and $(x_2, y_2, 0)$ are defined as two points on the calibration plane $Z = 0$ and $(x'_1, y'_1, 40)$ and $(x'_2, y'_2, 40)$ are two other points on the plane $Z = 40$. The line equations that pass each of the two points can be expressed by the following equations:

$$\vec{P} = (x_1, y_1, z_1) + t(x'_1 - x_1, y'_1 - y_1, 40) \quad (4)$$

$$\vec{Q} = (x_2, y_2, z_2) + t(x'_2 - x_2, y'_2 - y_2, 40) \quad (5)$$

$$\vec{P} = \vec{Q} \quad (6)$$

Since the equations given by Eq.(1) and Eq.(2) meet at the point C, i.e., Eq.(3), this point can be considered as the perspective center of the image, as shown in Fig. 3.

By using the perspective center of an image, the estimation of the image coordinates of any 3D world point P can be obtained. In this case, an ANN that is trained with the real world coordinates of points on $Z=0$ as inputs and the image plane coordinates for the corresponding points as targets is given. It should be noted that the input and target for the ANN in this case are different from those of the back-projection problem. When the image coordinate of a point (P_1) on any calibration plane Z is needed, the line equation that passes the point(P_1) in the calibration plane Z and the perspective center of a camera(C) is first obtained. The line equation can produce P_0 on the calibration plane $Z = 0$. By

Table 1. The specification of image acquisition

Image acquisition tool	Specification
Frame grabber	Horizontal resolution (X-axis) 512 Vertical resolution 512
CCD Image censor	Scale of cell(X-axis) $17 \mu m$ Scale of cell (Y-axis) $13 \mu m$
Lens	Focal length (F 1.4) 16mm

using P_0 as the input to the trained ANN, we can obtain the image coordinates of the point $\hat{p}$. This process is shown in Fig. 3.

3.2. Neural Network Implementation

The ANN model adopted in this paper is a standard MultiLayer Perceptron Type Neural Network (MLPNN) and an error back-propagation algorithm is used for training the MLPNN. After several experiments, the architecture of the MLPNN is selected as $2 \times 10 \times 8 \times 2$. Note that the selection of a specific architecture is a state of art and other architectures can be also used without any degradation of the resulting performance. With the architecture chosen, no over-fitting problem was experienced with 5,000 training epochs. Note that proper numbers of training epochs are dependent on the complexity of the given problem and the number of training data. Note that the neurons in the input and output layers represent the 2D coordinates. More detailed information on the MLPNN and error back-propagation algorithm can be found in [5].

Unlike the explicit camera calibration method, the proposed ANN-based method finds the direct relation between the world coordinates and the image coordinates. The ANN adopted in this implicit calibration approach can incorporate all the extrinsic parameters of the camera and the distortion parameters when the ANN is trained properly.

4. Experimental Results

4.1. Experimental Environment

The specifications of the image acquisition tool for our simulation environment are summarized in Table 1.

Images are acquired at three different positions. The performance of camera calibration results using artificial neural networks is compared and analyzed with that of Tsai's two stage method, the most widely used approach for explicit camera calibration. In this paper, the average error between the calibrated image coordinates and real world coordinates

is used to compare the performance of the camera calibration methods. The average error in pixels (AEIP) is defined as follows:

$$AEIP = \frac{1}{N} \sum_{i=1}^N [(X_{fi} - \hat{X}_{fi})^2 + (Y_{fi} - \hat{Y}_{fi})^2]^{1/2} \quad (7)$$

where $(\hat{X}_{fi}, \hat{Y}_{fi})$ is the estimated image coordinate, which is computed by using calibrated variables from the real coordinate point (X_{wi}, Y_{wi}, Z_{wi}) corresponding to the computer image coordinate (X_{fi}, Y_{fi}) .

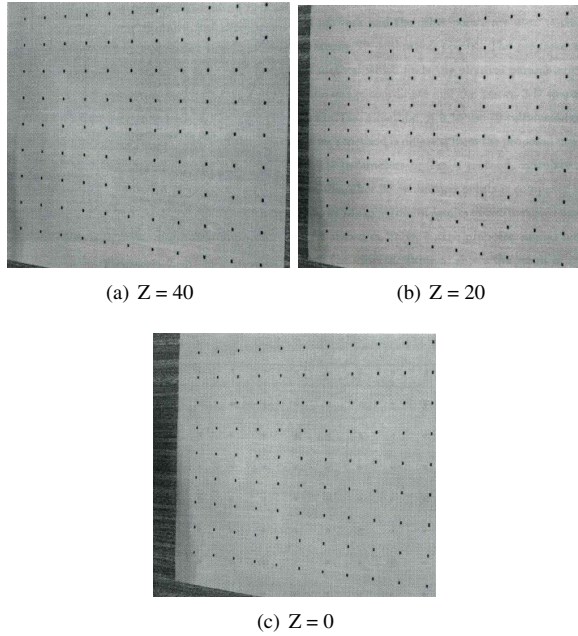


Figure 4. The calibration points at three different heights

The images used for the experiments are obtained by positioning the camera in the real world coordinate. The positions of the camera are also changed along the Z-axis for obtaining image data with different heights. Each image is composed of 99 calibration points (11×9) , which have an interval of 25mm between columns and an interval of 20mm between rows. Among the calibration points acquired from two images including 99 calibration points for each different heights, 79 randomly selected calibration points in each image are used for training the ANN and the remaining 20 points are used for evaluation of the trained ANN. Fig. 4 shows the images with different heights used in our experiments.

The proposed method is compared with Tsai's two stages method, which finds the physical parameters of the camera

Table 2. Comparison of estimation errors in AEIP

Case	Tsai's TSM	ANN-based
Case #1	0.5322	0.4829
Case #2	0.5277	0.4936
Case #3	0.5576	0.4642
Case #4	0.6201	0.5014
Case #5	0.5970	0.5235
Case #6	0.5128	0.4726
Case #7	0.6026	0.5198
Case #8	0.5844	0.4976
Case #9	0.6214	0.5527
Case #10	0.5993	0.5878
Average	0.5755	0.5096

Table 3. 3D world coordinate reconstruction error

World coordinate Z=20	Result of using ANN	Error
(75,0)	(74.8411, 0.6174)	0.6375
(175,0)	(174.3714, 0.3128)	0.7021
(150,60)	(150.3574, 59.8253)	0.3978
(100,80)	(99.1724, 81.0134)	1.3084
(175,100)	(174.9242, 99.4123)	0.5926
(255,100)	(224.9027, 100.0124)	0.0981
(25,120)	(24.6350, 120.3124)	0.4804
(150,140)	(149.9113, 139.0824)	0.9219
(125,160)	(126.0524, 160.3122)	1.0975
(175,160)	(175.2814, 159.8412)	0.3231
Average error	0.6559	

using the interrelation between the image coordinates and the known 3D space coordinates. For the calculation of the physical parameters for Tsai's method and training ANN, 10 sets of 79 randomly chosen calibration points are collected. For each set of calibration points, the remaining 20 points are used for testing the performance of both methods. Table 2 shows the test results for both methods. As shown in Table 2, the average improvement of the proposed ANN-based method over Tsai's method in terms of AEIP is 11.45 %.

4.2. Experiments on 3D Coordinate Reconstruction

The real space coordinate obtained by estimating the 3D space coordinate at an arbitrary height can be reconstructed after training the ANN with points on two calibration plans, i.e., $Z = 0$ and $Z = 40$, as follows: select a certain point

Table 4. Summary of estimation errors with real world coordinates at z=20

2D coordinates of image world	Average AEIP
at $Z = 0$	0.7120
at $Z = 40$	0.6826

of the image and then find the point of the real space coordinate of $Z=0$ and $Z=40$ calibration plane corresponding to the selected image point. Using Eq. (1) - Eq. (3), the perspective center of the image can be found. In our experiments, the coordinate of the perspective center is found as $C_x = 556.1233$, $C_y = 53.0954$, $C_z = 634.2666$. When the ANN is trained, ten points of an image are randomly selected first. The real points for the space coordinate of the $Z = 0$ and $Z = 40$ calibration plane corresponding to the selected image points are then found. By using Eq. (1) - Eq. (3), ten linear equations connecting points of $Z = 0$ plane with points of the $Z = 40$ plane are formulated for estimating the coordinate of the 3D space on the $Z = 20$ calibration plane. Table 3 shows the estimation results for 3D space.

4.3. Error Analysis

In order to estimate the 2D image coordinate, the center of perspective projection is first obtained. Ten arbitrary points on the $Z = 20$ plane are selected and the linear equations that connect the selected points with the obtained perspective center in the previous experiment are then found. Finally, intersection points on the $Z = 0$ and $Z = 40$ plane are obtained. Therefore, by using the trained ANN, the image coordinates from the intersecting points of the $Z = 0$ and $Z = 40$ plane are estimated. These experiments are performed with 10 randomly chosen training/test data sets. Table 4 shows the average results over 10 different sets of training/test sets on the 10 data points for coordinates of the 2D image from the intersecting points of the $Z = 0$ calibration plane and the $Z = 40$ calibration plane.

5. Conclusion

A camera calibration method using an artificial neural network is proposed in this paper. The proposed method calibrates a camera using the trained ANN instead of computing the physical camera parameters. The proposed ANN-based implicit method is applied to the estimation of 2D coordinates of an image world with given 3D space coordinates. The proposed method has advantages over Tsai's two stage method in real-time applications as it can be operated in real time after proper training while Tsai's two

stage method requires somewhat time consuming procedures for calculating proper parameters for a given task. The proposed method is also more flexible than Tsai's two stage method since it is not affected by camera position, illumination or distortion of the camera lens. More importantly, the proposed ANN-based method is not affected by the quality of the camera lens in finding the mapping function between the image coordinates and the real coordinates whereas Tsai's method is considerably affected by the quality of the camera. In comparison to the conventional approach Tsai's two stage method, the proposed ANN-based method shows promising results for calibrating camera when issues including practical applicability, flexibility, and real-time operation are relevant.

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